

Dry Eye Disease Risk Prediction Using Machine Learning

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Abstract

Dry Eye Disease (DED) is a common ocular condition that negatively affects quality of life and visual performance. The objective of this project was to develop a machine learning model capable of predicting Dry Eye Disease risk using demographic, lifestyle, medical, and eye-related factors. A dataset containing 20,000 records and 26 original features was analyzed. Data cleaning confirmed the absence of missing values and duplicate records. Feature engineering was applied to generate additional variables including BMI, Screen-Sleep Ratio, and Lifestyle Risk Score. Exploratory data analysis was performed to understand data distributions and relationships among variables. Feature selection identified the most informative predictors of Dry Eye Disease. Three machine learning algorithms, Logistic Regression, Random Forest, and Gradient Boosting, were trained and evaluated. Gradient Boosting achieved the highest accuracy of 70.05% and was selected as the final model. The model was deployed through a Streamlit web application, allowing users to predict Dry Eye Disease risk based on personal information. The results demonstrate the usefulness of machine learning techniques in supporting early identification of individuals at risk of Dry Eye Disease.

1. Introduction

Dry Eye Disease is a multifactorial condition characterized by insufficient tear production or excessive tear evaporation, leading to discomfort, visual disturbances, and ocular surface damage. Modern lifestyle factors such as prolonged screen exposure, inadequate sleep, and unhealthy habits have increased the prevalence of Dry Eye Disease worldwide. Early identification of individuals at risk can help reduce complications and improve quality of life. Machine learning offers powerful methods for analyzing large datasets and identifying patterns associated with disease occurrence.

This report presents the complete methodology and results of a machine learning project aimed at predicting Dry Eye Disease risk. The project covers data preprocessing, feature engineering, exploratory data analysis, model development, evaluation, and deployment.

2. Materials and Methods

2.1 Dataset

The dataset consisted of 20,000 observations with 26 original features describing demographic characteristics, sleep patterns, lifestyle habits, medical conditions, and eye symptoms. The target variable was a binary indicator of Dry Eye Disease presence.

2.2 Data Preprocessing

Data preprocessing included verifying dataset dimensions and checking for missing values, duplicate records, and incorrect data types. The dataset was found to be complete, with zero missing values and zero duplicate records. Blood pressure values, originally stored as a combined string, were separated into systolic and diastolic measurements to enable numerical analysis.

2.3 Feature Engineering

Three new features were constructed to enrich the predictive information in the dataset:

- Screen-Sleep Ratio: the ratio of average daily screen time to sleep duration, capturing imbalance between digital exposure and recovery.
- BMI: Body Mass Index calculated from weight and height measurements as a standard health indicator.
- Lifestyle Risk Score: a composite variable combining stress level, average screen time, and sleep quality to summarize cumulative lifestyle-related risk.

2.4 Exploratory Data Analysis

Exploratory data analysis was conducted using distribution plots, boxplots, and correlation heatmaps to understand the structure of the data and relationships between features and the target variable.

2.5 Feature Selection

Feature selection was performed using the SelectKBest method with mutual information scores. The top 10 features were identified as the most informative predictors of Dry Eye Disease.

2.6 Model Training and Evaluation

The dataset was divided into training and testing sets using an 80:20 split. Three classification algorithms were trained and evaluated: Logistic Regression, Random Forest, and Gradient Boosting. Model performance was assessed using classification accuracy, confusion matrices, and ROC-AUC curves.

3. Results and Discussion

3.1 Data Quality

The dataset was found to be complete without missing values or duplicate records, eliminating the need for imputation or deduplication. Data types were appropriate and no structural inconsistencies were identified. Blood pressure was successfully separated into systolic and diastolic components.

3.2 Exploratory Data Analysis

Figure 1 shows the distribution of Dry Eye Disease across the dataset. Disease cases were more prevalent than non-disease cases, indicating class imbalance. Figure 2 presents the age distribution of patients, showing a relatively uniform spread across age groups. Figure 3 illustrates the BMI distribution, which follows an approximately normal distribution centered around a BMI of 25.

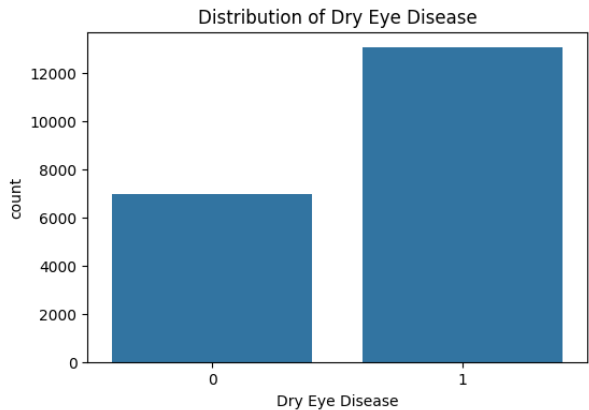


Figure 1. Distribution of Dry Eye Disease in the dataset (0 = No, 1 = Yes).

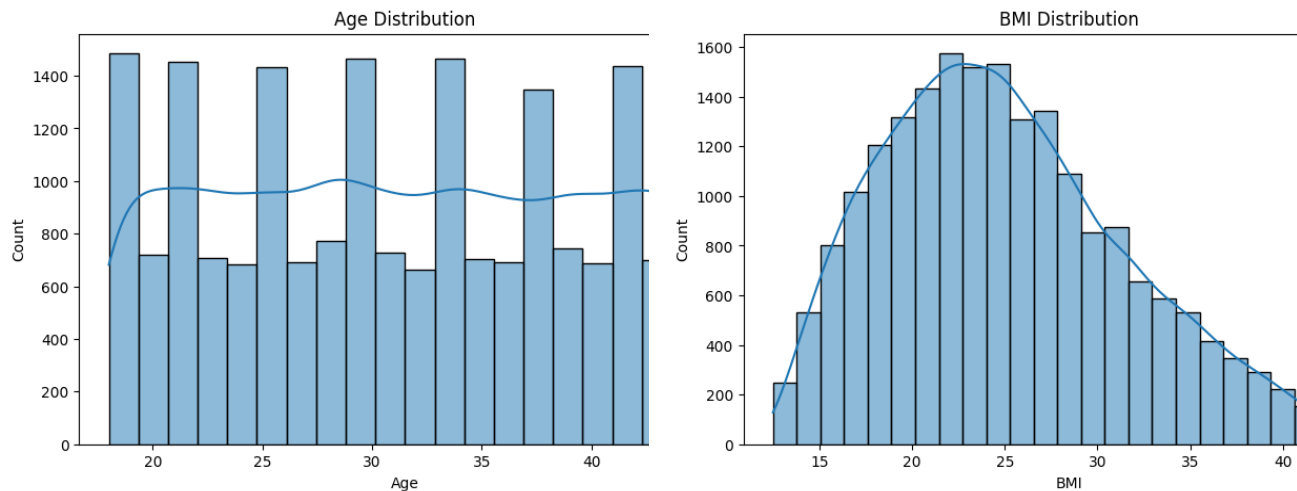


Figure 2. Age distribution of patients.

Figure 3. BMI distribution of patients.

Figures 4 and 5 present boxplot comparisons of screen time and stress level between disease and non-disease groups. Screen time was noticeably higher in individuals with Dry Eye Disease, while stress levels also differed between groups.

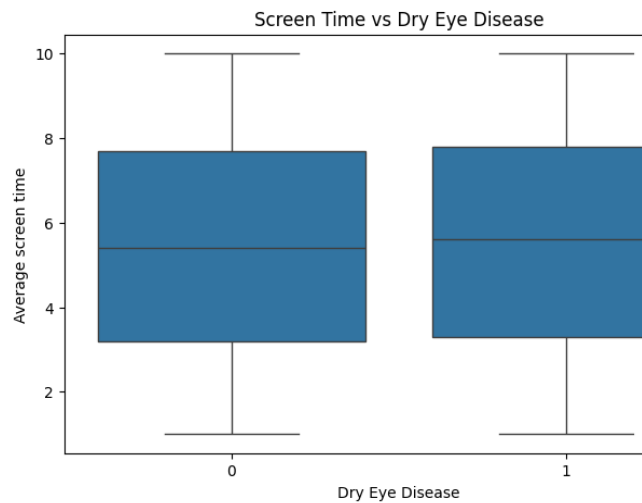


Figure 4. Screen time distribution by Dry Eye Disease status.



Figure 5. Stress level distribution by Dry Eye Disease status.

3.3 Correlation Analysis

Figure 6 presents the correlation heatmap of all features. Correlation analysis showed that screen-related factors and eye symptoms were positively associated with Dry Eye Disease occurrence. Specifically, eye-strain, redness, and itchiness showed meaningful correlation with the target variable. The Lifestyle Risk Score also demonstrated a notable relationship with disease status.

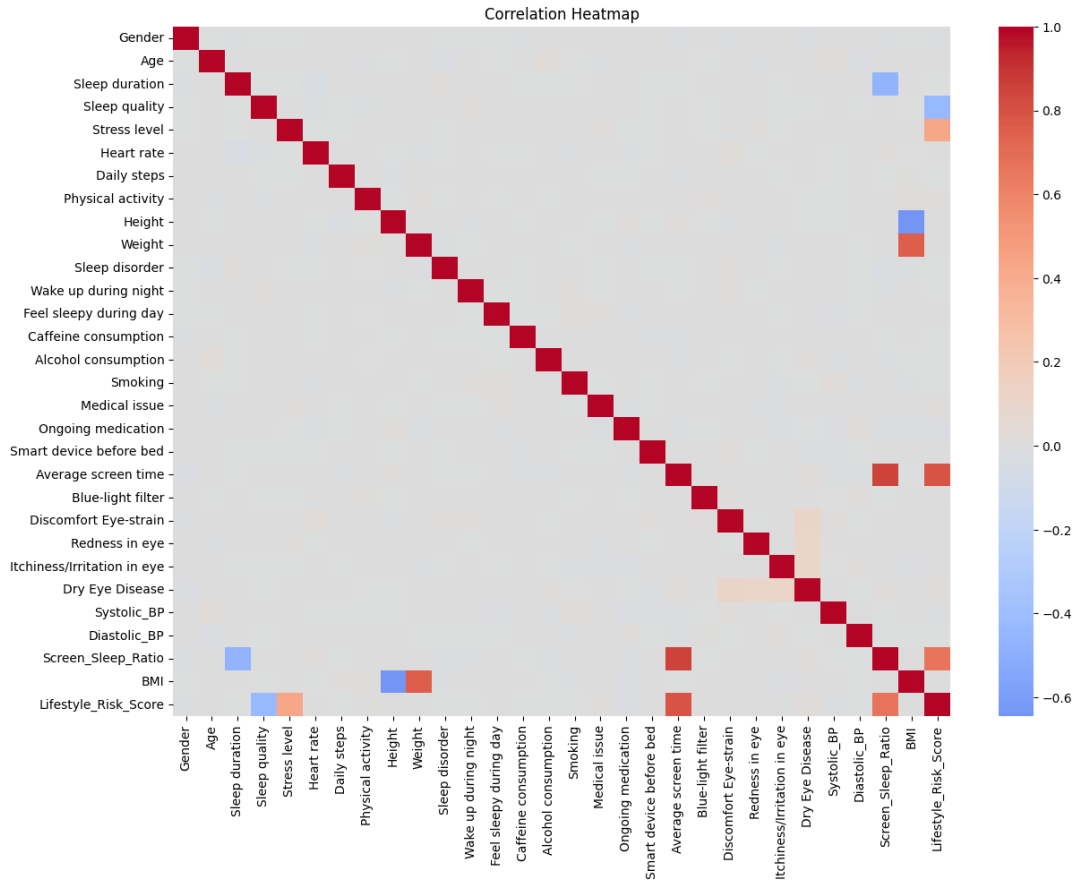


Figure 6. Correlation heatmap of all dataset features.

3.4 Feature Selection

Feature selection using mutual information identified the following ten features as the most informative predictors of Dry Eye Disease:

- Gender
- Daily Steps
- Smoking
- Smart Device Before Bed
- Average Screen Time
- Blue-Light Filter usage
- Discomfort / Eye-Strain
- Redness in Eye
- Itchiness / Irritation in Eye
- Diastolic Blood Pressure

The selected features reflect both lifestyle-related factors (screen time, smoking, device usage) and direct eye symptoms (redness, itchiness, strain), confirming the multifactorial nature of Dry Eye Disease.

3.5 Model Performance

Table 1 summarizes the classification accuracy of the three models evaluated. Gradient Boosting achieved the highest accuracy of 70.05%, followed by Random Forest (69.68%) and Logistic Regression (69.15%). Although the differences are small, Gradient Boosting consistently outperformed the other models and was selected as the final model.

Table 1. Classification accuracy of evaluated models.

Model	Accuracy	Rank
Logistic Regression	69.15%	3rd
Random Forest	69.68%	2nd
Gradient Boosting ★	70.05%	1st

Figure 7 provides a visual comparison of model accuracies. Figure 8 presents the confusion matrix for the Gradient Boosting model, showing the distribution of correct and incorrect predictions. Figure 9 shows the ROC curve, with an AUC score of 0.571, indicating modest discriminative ability above random chance.

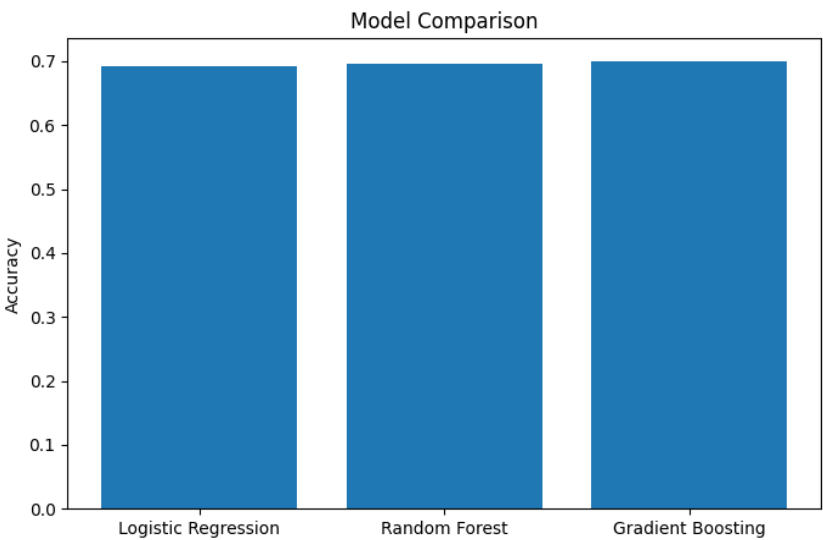


Figure 7. Comparison of model classification accuracies.

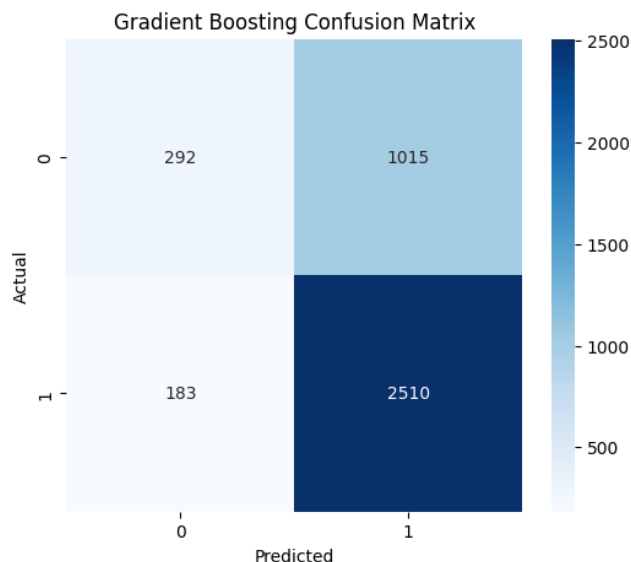


Figure 8. Confusion matrix for the Gradient Boosting model.

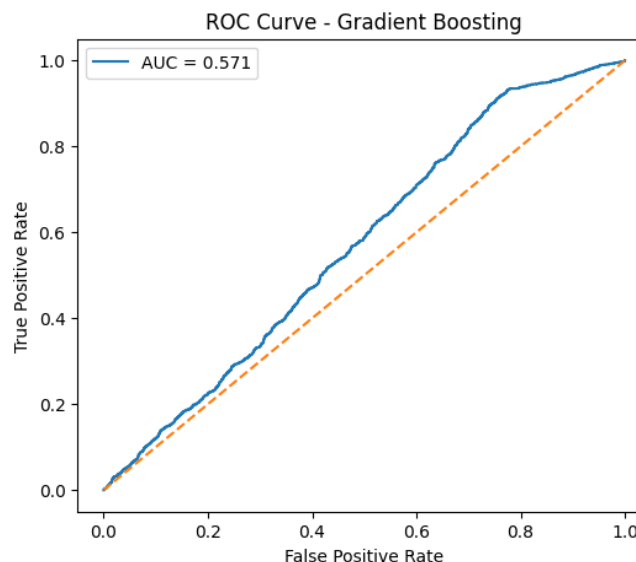


Figure 9. ROC curve for the Gradient Boosting model (AUC = 0.571).

3.6 Learning Curve

Figure 10 presents the learning curve for the Gradient Boosting model. The training and cross-validation scores converge as the training set size increases, indicating that the model generalizes reasonably well and does not exhibit severe overfitting.

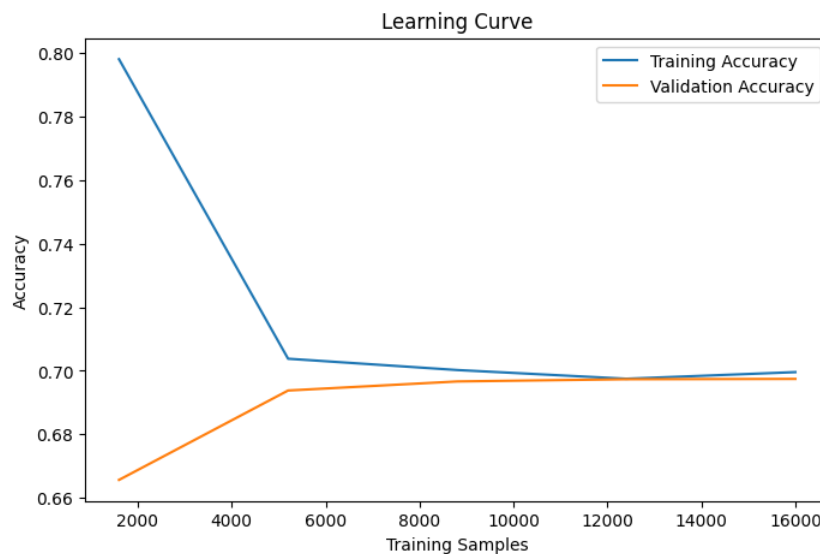


Figure 10. Learning curve of the Gradient Boosting model.

3.7 Deployment

The final Gradient Boosting model was exported and integrated into a Streamlit web application. The application provides an interactive interface where users can input their personal information and

receive an immediate Dry Eye Disease risk prediction. This tool makes the predictive model accessible to non-technical users without requiring any programming knowledge.

4. Conclusion

This study demonstrated the successful application of machine learning techniques for predicting Dry Eye Disease risk from demographic, lifestyle, and clinical data. Feature engineering and selection improved the quality of the predictive analysis by focusing on the most informative variables. Among the three evaluated algorithms, Gradient Boosting provided the best overall classification accuracy of 70.05%.

The developed web application offers a simple and accessible tool for estimating Dry Eye Disease risk based on user-provided information. Future work may involve larger and more diverse datasets, additional clinical features, class balancing techniques, and advanced machine learning approaches such as ensemble stacking or deep learning to further improve predictive accuracy and clinical utility.